

SMB Authentication & NTLM Logon Detection using Wazuh

Objective

The objective of this simulation was to investigate how Windows Security Events and Wazuh detect failed SMB authentication attempts, successful NTLM-based network logons, privileged account authentication, and SMB activity commonly associated with lateral movement and credential-based attacks.

MITRE ATT&CK Mapping

Tactic	Technique	ID
Credential Access	Brute Force	T1110
Credential Access	Valid Accounts	T1078
Discovery	Account Discovery	T1087
Lateral Movement	SMB/Windows Admin Shares	T1021.002
Lateral Movement	Remote Services	T1021

Attack Simulation

The following SMB enumeration command was executed from the Kali Linux system against the monitored Windows 11 endpoint:

```
smbclient -L //192.168.56.106/ -U socadmin
```

The simulation intentionally generated:

- failed SMB authentication attempts using incorrect credentials
- successful NTLM network logon activity using valid credentials

This approach reproduced realistic SMB authentication and enumeration behavior commonly observed during reconnaissance and lateral movement activity.

Attack Chain Overview

Step	Activity
1	SMB authentication initiated from Kali Linux
2	Failed NTLM authentication generated
3	Successful NTLMv2 network logon established
4	Administrative privileges assigned
5	SMB session terminated

```
(kali@kali)-[~]
$ smbclient -L //192.168.56.106/ -U socadmin
Password for [WORKGROUP\socadmin]:
session setup failed: NT_STATUS_LOGON_FAILURE

(kali@kali)-[~]
$ smbclient -L //192.168.56.106/ -U socadmin
Password for [WORKGROUP\socadmin]:

Sharename      Type           Comment
-----
ADMIN$         Disk          Remote Admin
C$             Disk          Default share
IPC$           IPC           Remote IPC
Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to 192.168.56.106 failed (Error NT_STATUS_RESOURCE_NAM
E_NOT_FOUND)
Unable to connect with SMB1 -- no workgroup available

(kali@kali)-[~]
$
```

Figure 1 — SMB authentication attempts initiated from Kali Linux system

FAILED SMB AUTHENTICATION ANALYSIS

Key Security Events Identified

Event ID / Rule ID	Detection Purpose
4625	Failed SMB authentication
4624	Successful SMB network logon
4672	Special privileges assigned
4634	SMB session logoff
60122	Wazuh failed authentication detection
92657	Wazuh suspicious remote NTLM logon detection

Windows Event ID 4625 Analysis

Key Findings

Field	Value
Event ID	4625
Logon Type	3
Target Username	socadmin
Source IP	192.168.56.107
Workstation	KALI
Authentication Package	NTLM
SubStatus	0xc000006a

Investigation Analysis

Windows Security Event ID 4625 captured a failed SMB network authentication attempt originating from the Kali Linux system.

The authentication request targeted the account:

socadmin

The authentication failed because an incorrect password was supplied for a valid account.

The following SubStatus value was observed:

0xc000006a

Meaning:

valid username, incorrect password

This behavior is operationally important because it closely resembles:

- password spraying
- credential guessing
- brute-force authentication attempts
- lateral movement authentication failures

The investigation also confirmed:

- remote SMB authentication activity
- NTLM authentication usage
- external source attribution
- Logon Type 3 network authentication behavior

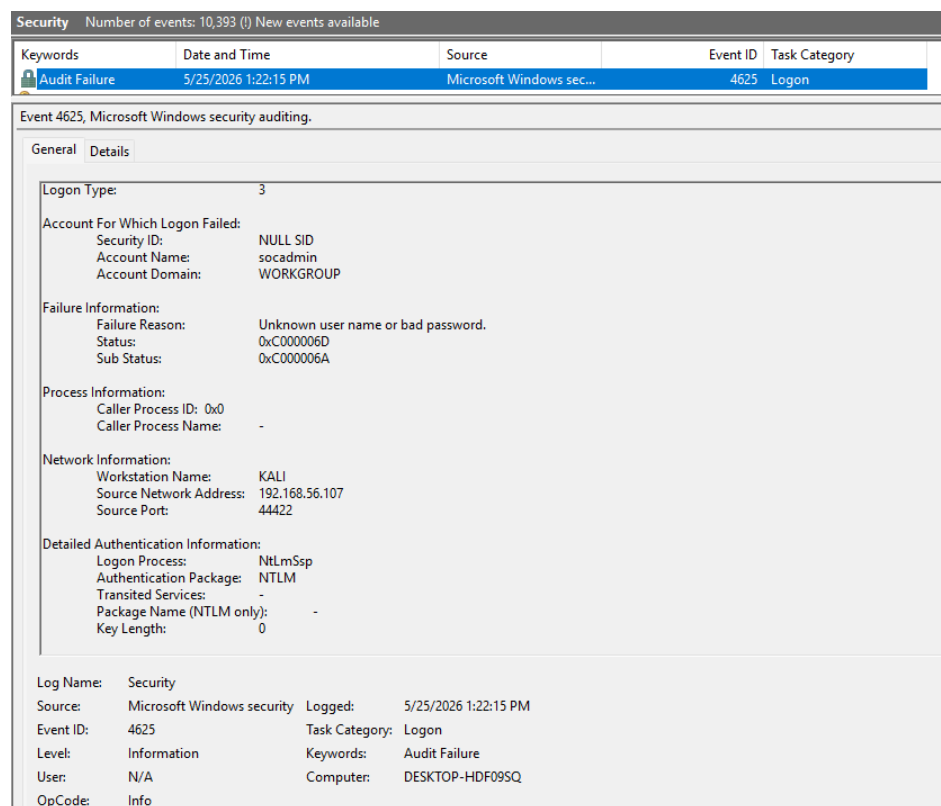


Figure 2 — Windows Event ID 4625 showing failed SMB authentication attempt against valid account

SUCCESSFUL SMB NETWORK LOGON ANALYSIS

Windows Event ID 4624 Analysis

Key Findings

Field	Value
Event ID	4624
Logon Type	3
Account Name	socadmin
Source IP	192.168.56.107
Workstation	KALI
Authentication Package	NTLM
NTLM Version	NTLM V2

Investigation Analysis

Following the failed authentication attempt, a successful SMB network logon was established using valid credentials.

Windows Security Event ID 4624 recorded a successful remote NTLMv2 authentication originating from the Kali Linux system.

The successful authentication established:

- remote SMB access
- authenticated NTLMv2 session
- external source attribution
- network-based administrative authentication

This event was highly valuable because successful remote NTLM authentication commonly appears during:

- lateral movement
- administrative share access
- remote service activity
- credential abuse investigations

The use of NTLMv2 also demonstrated modern SMB authentication behavior rather than legacy NTLMv1 negotiation.

IMPORTANT ANALYST NOTE

Successful SMB authentication events should always be investigated together with preceding authentication failures, source IP activity, and privilege assignment events to determine whether the behavior represents legitimate administration or suspicious lateral movement activity.

Security Number of events: 10,393 (0) New events available

Keywords	Date and Time	Source	Event ID	Task Category
Audit Success	5/25/2026 1:22:34 PM	Microsoft Windows sec...	4624	Logon

Event 4624, Microsoft Windows security auditing.

General Details

Logon information:

- Logon Type: 3
- Restricted Admin Mode: -
- Remote Credential Guard: -
- Virtual Account: No
- Elevated Token: No

Impersonation Level: Impersonation

New Logon:

- Security ID: DESKTOP-HDF09SQ\socadmin
- Account Name: socadmin
- Account Domain: DESKTOP-HDF09SQ
- Logon ID: 0xC8E2CF
- Linked Logon ID: 0x0
- Network Account Name: -
- Network Account Domain: -
- Logon GUID: {00000000-0000-0000-0000-000000000000}

Process Information:

- Process ID: 0x0
- Process Name: -

Network Information:

- Workstation Name: KALI
- Source Network Address: 192.168.56.107
- Source Port: 32962

Log Name: Security

Source: Microsoft Windows security Logged: 5/25/2026 1:22:34 PM

Event ID: 4624 Task Category: Logon

Level: Information Keywords: Audit Success

User: N/A Computer: DESKTOP-HDF09SQ

OpCode: Info

Detailed Authentication Information:

- Logon Process: NtLmSsp
- Authentication Package: NTLM
- Transited Services: -
- Package Name (NTLM only): NTLM V2
- Key Length: 128

This event is generated when a logon session is created. It is generated on the computer that was accessed.

The subject fields indicate the account on the local system which requested the logon. This is most commonly a service such as the Server service, or a local process such as Winlogon.exe or Services.exe.

The logon type field indicates the kind of logon that occurred. The most common types are 2 (interactive) and 3 (network).

The New Logon fields indicate the account for whom the new logon was created, i.e. the account that was logged on.

The network fields indicate where a remote logon request originated. Workstation name is not always available and may be left blank in some cases.

The impersonation level field indicates the extent to which a process in the logon session can impersonate.

The authentication information fields provide detailed information about this specific logon request.

- Logon GUID is a unique identifier that can be used to correlate this event with a KDC event.
- Transited services indicate which intermediate services have participated in this logon request.
- Package name indicates which sub-protocol was used among the NTLM protocols.
- Key length indicates the length of the generated session key. This will be 0 if no session key was requested.

Figure 3 — Windows Event ID 4624 showing successful remote SMB NTLMv2 authentication

PRIVILEGE ASSIGNMENT ANALYSIS

Windows Event ID 4672 Analysis

Key Findings

Field	Value
Event ID	4672
Username	socadmin
Logon ID	0xC8E2CF
Privilege Level	Administrative

Critical Privileges Observed

Privilege	Meaning
SeDebugPrivilege	Debug or access protected processes
SeBackupPrivilege	Access protected files
SeRestorePrivilege	Restore protected system data
SeTakeOwnershipPrivilege	Take ownership of objects
SeImpersonatePrivilege	Token impersonation capability

Investigation Analysis

Windows Event ID 4672 recorded the assignment of elevated administrative privileges immediately after successful SMB authentication.

This telemetry significantly increased detection confidence because privileged logons following remote network authentication are commonly associated with lateral movement, remote administration, and post-compromise activity.

The assigned privileges indicated that the authenticated account possessed administrative-level access capable of:

- privileged system interaction
- administrative SMB access
- remote management activity
- elevated process interaction

Although the activity originated from an authorized lab environment, the behavioral pattern closely resembled real-world privileged authentication and lateral movement activity frequently observed during enterprise SOC investigations.

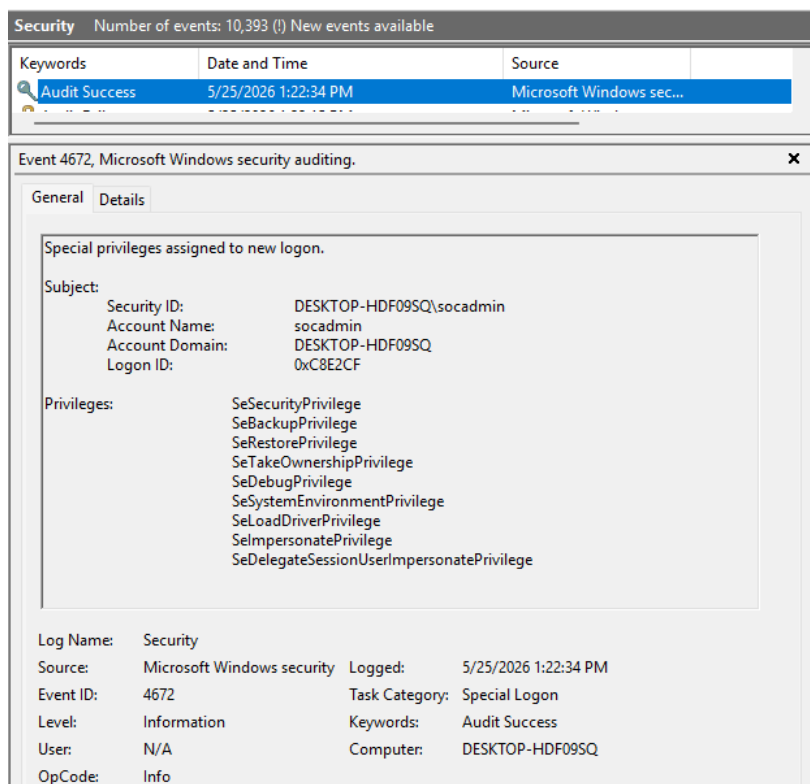


Figure 4 — Windows Event ID 4672 showing administrative privilege assignment after SMB authentication

Windows Event ID 4634 Analysis

Key Findings

Field	Value
Event ID	4634
Account Name	socadmin
Logon Type	3
Source IP	192.168.56.107

Investigation Analysis

Windows Event ID 4634 recorded termination of the SMB authentication session following successful network logon activity.

The logoff event completed the authentication timeline and confirmed that the SMB session was short-lived and directly associated with network authentication activity originating from the Kali Linux system.

Correlation between Event IDs 4625, 4624, and 4634 provided a complete view of the authentication lifecycle, including failed access attempts, successful NTLM authentication, and session termination behavior.

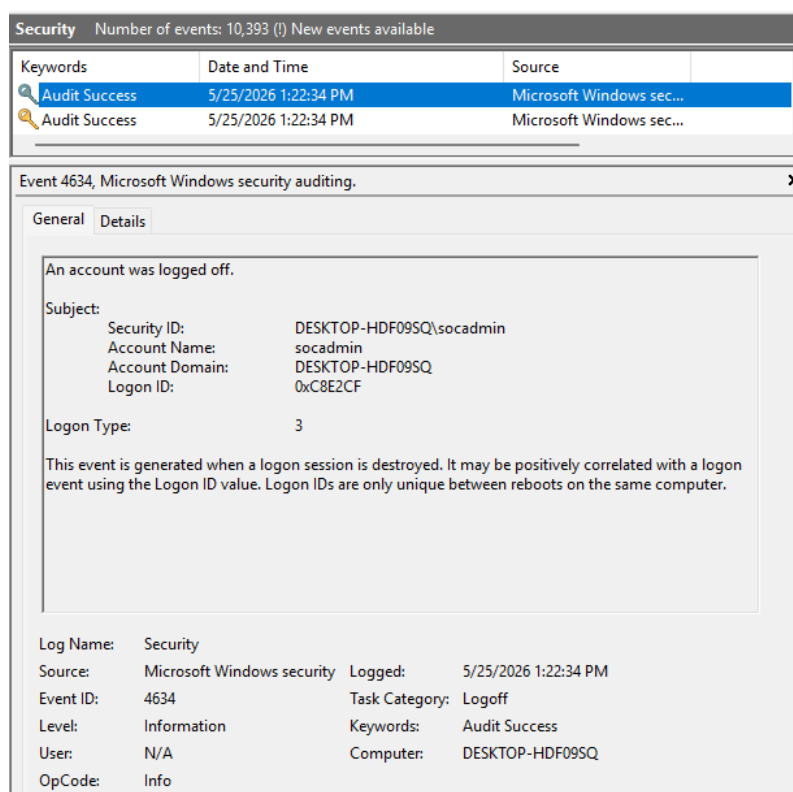


Figure 5 — Windows Event ID 4634 showing SMB network session termination

WAZUH DETECTION ANALYSIS

Wazuh Rules Triggered

Rule ID	Severity	Description
60122	5	Logon Failure - Unknown user or bad password
92657	6	Successful Remote NTLM Logon Detected
60137	3	Windows User Logoff
67028	3	Special privileges assigned to new logon

Investigation Analysis

Wazuh successfully correlated multiple authentication-related events generated during SMB enumeration and NTLM-based network logon activity.

Analysis of the alert sequence identified:

- failed SMB authentication attempts
- successful NTLMv2 network logons
- administrative privilege assignment
- remote source attribution
- SMB session lifecycle activity

Behavioral correlation across Windows Security Events and Wazuh alerts increased overall detection confidence and provided clearer visibility into the authentication workflow originating from the Kali Linux system.

The successful NTLM authentication event was additionally associated with:

- potential lateral movement behavior
- pass-the-hash style authentication patterns
- remote administrative access activity

This highlights how SIEM platforms leverage event correlation and behavioral analysis rather than relying solely on isolated authentication events.

IMPORTANT FALSE POSITIVE CONSIDERATIONS

Although the activity originated from authorized lab testing, the overall authentication pattern closely resembled reconnaissance and credential-based lateral movement behavior commonly observed in enterprise environments.

Analyst validation was therefore critical to distinguish legitimate security testing activity from potentially malicious SMB authentication behavior and unauthorized remote access attempts.

```
May 25, 2026 @ 13:22:32.108 input.type: log agent.ip: 10.0.2.15 agent.name: DESKTOP-HDF09SQ agent.id: 001 manager.name: wazuh-server
data.win.eventdata.subjectLogonId: 0x0 data.win.eventdata.ipAddress: 192.168.56.107
data.win.eventdata.authenticationPackageName: NTLM data.win.eventdata.workstationName: KALI data.win.eventdata.subStatus: 0xc0
00006a data.win.eventdata.logonProcessName: NtLmSsp data.win.eventdata.targetUserName: socadmin
data.win.eventdata.keyLength: 0 data.win.eventdata.subjectUserSid: S-1-0-0 data.win.eventdata.processId: 0x0
```

```
{
  "_index": "wazuh-alerts-4.x-2026.05.25",
  "_id": "xjYfXp4Br749Nx0oE9Y0",
  "_version": 1,
  "_score": null,
  "_source": {
    "input": {
      "type": "log"
    },
    "agent": {
      "ip": "10.0.2.15",
      "name": "DESKTOP-HDF09SQ",
      "id": "001"
    },
    "manager": {
      "name": "wazuh-server"
    },
    "data": {
      "win": {
        "eventdata": {
          "subjectLogonId": "0x0",
          "ipAddress": "192.168.56.107",
          "authenticationPackageName": "NTLM",
          "workstationName": "KALI",

```



```

"subStatus": "0xc000006a",
"logonProcessName": "NtLmSsp",
"targetUserName": "socadmin",
"keyLength": "0",
"subjectUserSid": "S-1-0-0",
"processId": "0x0",
"ipPort": "44422",
"failureReason": "%%2313",
"targetDomainName": "WORKGROU",
"targetUserSid": "S-1-0-0",
"logonType": "3",
"status": "0xc000006d"
},
"system": {
"eventID": "4625",
"keywords": "0x8010000000000000",
"providerGuid": "{54849625-5478-4994-a5ba-3e3b0328c30d}",
"level": "0",
"channel": "Security",
"opcode": "0",
"message": "\"An account failed to log on.\r\n\r\nSubject:\r\n\tSecurity ID:\t\tS-1-0-0\r\n\tAccount Name:\t\t-\r\n\tAccount Domain:\t\t-\r\n\tLogon ID:\t\t0x0\r\n\r\nLogon Type:\t\t3\r\n\r\nAccount For Which Logon\r\nFailed:\r\n\tSecurity ID:\t\tS-1-0-0\r\n\tAccount Name:\tsocadmin\r\n\tAccount Domain:\t\tWORKGROU\r\n\r\nFailure\r\nInformation:\r\n\tFailure Reason:\t\tUnknown user name or bad password.\r\n\tStatus:\t\t0xC000006D\r\n\tSub\r\nStatus:\t\t0xC000006A\r\n\r\nProcess Information:\r\n\tCaller Process ID:\t0x0\r\n\tCaller Process Name:\t-\r\n\r\n\r\nNetwork Information:\r\n\tWorkstation Name:\tKALI\r\n\tSource Network Address:\t192.168.56.107\r\n\tSource\r\nPort:\t\t44422\r\n\r\n\r\nDetailed Authentication Information:\r\n\tLogon Process:\t\tNtLmSsp \r\n\tAuthentication\r\nPackage:\tNTLM\r\n\tTransited Services:\t-\r\n\tPackage Name (NTLM only):\t-\r\n\tKey Length:\t\t0\r\n\r\n\r\nThis event is\r\ngenerated when a logon request fails. It is generated on the computer where access was attempted.\r\n\r\n\r\nThe Subject\r\nfields indicate the account on the local system which requested the logon. This is most commonly a service such as\r\nthe Server service, or a local process such as Winlogon.exe or Services.exe.\r\n\r\n\r\nThe Logon Type field indicates the\r\nkind of logon that was requested. The most common types are 2 (interactive) and 3 (network).\r\n\r\n\r\nThe Process\r\nInformation fields indicate which account and process on the system requested the logon.\r\n\r\n\r\nThe Network\r\nInformation fields indicate where a remote logon request originated. Workstation name is not always available and may\r\nbe left blank in some cases.\r\n\r\n\r\nThe authentication information fields provide detailed information about this specific\r\nlogon request.\r\n\r\n\t- Transited services indicate which intermediate services have participated in this logon\r\nrequest.\r\n\r\n\t- Package name indicates which sub-protocol was used among the NTLM protocols.\r\n\r\n\t- Key length\r\nindicates the length of the generated session key. This will be 0 if no session key was requested.\"\"",
"version": "0",
"systemTime": "2026-05-25T07:52:15.0389830Z",
"eventRecordID": "10389",
"threadID": "960",
"computer": "DESKTOP-HDF09SQ",
"task": "12544",
"processID": "868",
"severityValue": "AUDIT_FAILURE",
"providerName": "Microsoft-Windows-Security-Auditing"
}
},
"rule": {
"mail": false,
"level": 5,
"pci_dss": [
"10.2.4",
"10.2.5"
],
"hipaa": [

```

```

"164.312.b"
],
"tsc": [
  "CC6.1",
  "CC6.8",
  "CC7.2",
  "CC7.3"
],
"description": "Logon Failure - Unknown user or bad password",
"groups": [
  "windows",
  "windows_security",
  "authentication_failed"
],
"nist_800_53": [
  "AU.14",
  "AC.7"
],
"gdpr": [
  "IV_35.7.d",
  "IV_32.2"
],
"firedtimes": 2,
"mitre": {
  "technique": [
    "Account Access Removal"
  ],
  "id": [
    "T1531"
  ],
  "tactic": [
    "Impact"
  ]
},
"id": "60122",
"gpq13": [
  "7.1"
],
"location": "EventChannel",
"decoder": {
  "name": "windows_eventchannel"
},
"id": "1779695552.1901076",
"timestamp": "2026-05-25T07:52:32.100+0000",
"fields": {
  "timestamp": [
    "2026-05-25T07:52:32.100Z"
  ]
},
"highlight": {
  "rule.id": [
    "@opensearch-dashboards-highlighted-field@60122@/opensearch-dashboards-highlighted-field@"
  ]
},
"sort": [
  1779695552100
]

```

```
]
}
```

Figure 6 — Wazuh detection for failed SMB authentication attempt

```
May 25, 2026 @ 13:22:44.910 input.type: log agent.ip: 10.0.2.15 agent.name: DESKTOP-HDF09SQ agent.id: 001 manager.name: wazuh-server
data.win.eventdata.subjectLogonId: 0x0 data.win.eventdata.targetLinkedLogonId: 0x0 data.win.eventdata.impersonationLevel: %18
33 data.win.eventdata.ipAddress: 192.168.56.107 data.win.eventdata.authenticationPackageName: NTLM
data.win.eventdata.workstationName: KALI data.win.eventdata.lmPackageName: NTLM V2 data.win.eventdata.targetLogonId: 0xc8e2cf
data.win.eventdata.logonProcessName: NtLmSsp data.win.eventdata.logonGuid: {00000000-0000-0000-0000-000000000000}

{
  "_index": "wazuh-alerts-4.x-2026.05.25",
  "_id": "yTYfXp4Br749Nx0oTdbO",
  "_version": 1,
  "_score": null,
  "_source": {
    "input": {
      "type": "log"
    },
    "agent": {
      "ip": "10.0.2.15",
      "name": "DESKTOP-HDF09SQ",
      "id": "001"
    },
    "manager": {
      "name": "wazuh-server"
    },
    "data": {
      "win": {
        "eventdata": {
          "subjectLogonId": "0x0",
          "targetLinkedLogonId": "0x0",
          "impersonationLevel": "%%1833",
          "ipAddress": "192.168.56.107",
          "authenticationPackageName": "NTLM",
          "workstationName": "KALI",
          "lmPackageName": "NTLM V2",
          "targetLogonId": "0xc8e2cf",
          "logonProcessName": "NtLmSsp",
          "logonGuid": "{00000000-0000-0000-0000-000000000000}",
          "targetUserName": "socadmin",
          "keyLength": "128",
          "elevatedToken": "%%1843",
          "subjectUserSid": "S-1-0-0",
          "processId": "0x0",
          "ipPort": "32962",
          "targetDomainName": "DESKTOP-HDF09SQ",
          "targetUserSid": "S-1-5-21-221611468-1794481200-2977160085-1001",
          "virtualAccount": "%%1843",
          "logonType": "3"
        },
        "system": {
          "eventId": "4624",
          "keywords": "0x8020000000000000",
          "providerGuid": "{54849625-5478-4994-a5ba-3e3b0328c30d}",
          "level": "0",
          "channel": "Security",
          "opcode": "0",
          "message": "\\An account was successfully logged on.\\r\\n\\r\\nSubject:\\r\\n\\tSecurity ID:\\t\\tS-1-0-0\\r\\n\\tAccount
```

Name:\t\t-\r\n\tAccount Domain:\t\t-\r\n\tLogon ID:\t\t0x0\r\n\r\nLogon Information:\r\n\tLogon Type:\t\t3\r\n\tRestricted Admin Mode:\t\t-\r\n\tRemote Credential Guard:\t\t-\r\n\tVirtual Account:\t\tNo\r\n\tElevated Token:\t\tNo\r\n\r\nImpersonation Level:\t\tImpersonation\r\n\r\nNew Logon:\r\n\tSecurity ID:\t\tS-1-5-21-221611468-1794481200-2977160085-1001\r\n\tAccount Name:\t\tsocadmin\r\n\tAccount Domain:\t\tDESKTOP-HDF09SQ\r\n\tLogon ID:\t\t0xC8E2CF\r\n\tLinked Logon ID:\t\t0x0\r\n\tNetwork Account Name:\t\t-\r\n\tNetwork Account Domain:\t\t-\r\n\tLogon GUID:\t\t{00000000-0000-0000-0000-000000000000}\r\n\r\nProcess Information:\r\n\tProcess ID:\t\t0x0\r\n\tProcess Name:\t\t-\r\n\r\nNetwork Information:\r\n\tWorkstation Name:\t\tKALI\r\n\tSource Network Address:\t\t192.168.56.107\r\n\tSource Port:\t\t32962\r\n\r\nDetailed Authentication Information:\r\n\tLogon Process:\t\tNtLmSsp \r\n\tAuthentication Package:\t\tNTLM\r\n\tTransited Services:\t\t-\r\n\tPackage Name (NTLM only):\t\tNTLM V2\r\n\tKey Length:\t\t128\r\n\r\nThis event is generated when a logon session is created. It is generated on the computer that was accessed.\r\n\r\nThe subject fields indicate the account on the local system which requested the logon. This is most commonly a service such as the Server service, or a local process such as Winlogon.exe or Services.exe.\r\n\r\nThe logon type field indicates the kind of logon that occurred. The most common types are 2 (interactive) and 3 (network).\r\n\r\nThe New Logon fields indicate the account for whom the new logon was created, i.e. the account that was logged on.\r\n\r\nThe network fields indicate where a remote logon request originated. Workstation name is not always available and may be left blank in some cases.\r\n\r\nThe impersonation level field indicates the extent to which a process in the logon session can impersonate.\r\n\r\nThe authentication information fields provide detailed information about this specific logon request.\r\n\t- Logon GUID is a unique identifier that can be used to correlate this event with a KDC event.\r\n\t- Transited services indicate which intermediate services have participated in this logon request.\r\n\t- Package name indicates which sub-protocol was used among the NTLM protocols.\r\n\t- Key length indicates the length of the generated session key. This will be 0 if no session key was requested.\t\t",

```

"version": "3",
"systemTime": "2026-05-25T07:52:34.1668885Z",
"eventRecordID": "10392",
"threadID": "3048",
"computer": "DESKTOP-HDF09SQ",
"task": "12544",
"processID": "868",
"severityValue": "AUDIT_SUCCESS",
"providerName": "Microsoft-Windows-Security-Auditing"
}
},
"rule": {
"mail": false,
"level": 6,
"pci_dss": [
"10.2.5"
],
"hipaa": [
"164.312.b"
],
"tsc": [
"CC6.8",
"CC7.2",
"CC7.3"
],
"description": "Successful Remote Logon Detected - User:\\socadmin - NTLM authentication, possible pass-the-hash attack - Possible RDP connection. Verify that KALI is allowed to perform RDP connections",
"groups": [
"win_evt_channel",
"windows",
"authentication_success"
],
"nist_800_53": [
"AU.14",
"AC.7"
]
}

```

```

],
"gdpr": [
"IV_32.2"
],
"firedtimes": 1,
"mitre": {
"technique": [
"Pass the Hash",
"Domain Accounts",
"Remote Desktop Protocol"
],
"id": [
"T1550.002",
"T1078.002",
"T1021.001"
],
"tactic": [
"Defense Evasion",
"Lateral Movement",
"Persistence",
"Privilege Escalation",
"Initial Access"
]
},
"id": "92657",
"gpg13": [
"7.1",
"7.2"
]
},
"location": "EventChannel",
"decoder": {
"name": "windows_eventchannel"
},
"id": "1779695564.1913381",
"full_log": "{\nwin\":{\"system\":{\"providerName\":\"Microsoft-Windows-Security-Auditing\",\"providerGuid\":\"{54849625-5478-4994-a5ba-3e3b0328c30d}\",\n\"eventID\":\"4624\", \"version\":\"3\", \"level\":\"0\", \"task\":\"12544\", \"opcode\":\"0\", \"keywords\":\"0x805-25T07:52:34.1668885Z\", \"eventRecordID\":\"10392\", \"processID\":\"868\", \"threadID\":\"3048\", \"channel\":\"Security\", \"HDF09SQ\", \"severityValue\":\"AUDIT_SUCCESS\", \"message\": \"\\\\\\\\\\\\\\\\An account was successfully logged on.\\\\\\\\\\\\\\\\\\\\n\\\\\\\\\\\\\\\\\\\\nSubject:\\\\\\\\\\\\\\\\\\\\nSecurity ID:\\\\\\\\\\\\\\\\\\\\nS-1-0-0\\\\\\\\\\\\\\\\\\\\nAccount Name:\\\\\\\\\\\\\\\\\\\\n\\\\\\\\\\\\\\\\\\\\nAccount Domain:\\\\\\\\\\\\\\\\\\\\n\\\\\\\\\\\\\\\\\\\\nLogon ID:\\\\\\\\\\\\\\\\\\\\n0x0\\\\\\\\\\\\\\\\\\\\nLogon Information:\\\\\\\\\\\\\\\\\\\\nLogon Type:\\\\\\\\\\\\\\\\\\\\n3\\\\\\\\\\\\\\\\\\\\nRestricted Admin Mode:\\\\\\\\\\\\\\\\\\\\n\\\\\\\\\\\\\\\\\\\\nRemote Credential Guard:\\\\\\\\\\\\\\\\\\\\nVirtual Account:\\\\\\\\\\\\\\\\\\\\nNo\\\\\\\\\\\\\\\\\\\\nElevated Token:\\\\\\\\\\\\\\\\\\\\nNo\\\\\\\\\\\\\\\\\\\\nImpersonation Level:\\\\\\\\\\\\\\\\\\\\nImpersonation\\\\\\\\\\\\\\\\\\\\nNew Logon:\\\\\\\\\\\\\\\\\\\\nSecurity ID:\\\\\\\\\\\\\\\\\\\\nS-1-5-21-221611468-1794481200-2977160085-1001\\\\\\\\\\\\\\\\\\\\nAccount Name:\\\\\\\\\\\\\\\\\\\\nsocadmin\\\\\\\\\\\\\\\\\\\\nAccount Domain:\\\\\\\\\\\\\\\\\\\\nDESKTOP-HDF09SQ\\\\\\\\\\\\\\\\\\\\nLogon ID:\\\\\\\\\\\\\\\\\\\\n0xC8E2CF\\\\\\\\\\\\\\\\\\\\nLinked Logon ID:\\\\\\\\\\\\\\\\\\\\n0x0\\\\\\\\\\\\\\\\\\\\nNetwork Account Name:\\\\\\\\\\\\\\\\\\\\n\\\\\\\\\\\\\\\\\\\\nNetwork Account Domain:\\\\\\\\\\\\\\\\\\\\n\\\\\\\\\\\\\\\\\\\\nLogon GUID:\\\\\\\\\\\\\\\\\\\\n{00000000-0000-0000-0000-000000000000}\\\\\\\\\\\\\\\\\\\\nProcess Information:\\\\\\\\\\\\\\\\\\\\nProcess ID:\\\\\\\\\\\\\\\\\\\\n0x0\\\\\\\\\\\\\\\\\\\\nProcess Name:\\\\\\\\\\\\\\\\\\\\n\\\\\\\\\\\\\\\\\\\\nNetwork Information:\\\\\\\\\\\\\\\\\\\\nWorkstation Name:\\\\\\\\\\\\\\\\\\\\nKALI\\\\\\\\\\\\\\\\\\\\nSource Network Address:\\\\\\\\\\\\\\\\\\\\n192.168.56.107\\\\\\\\\\\\\\\\\\\\nSource Port:\\\\\\\\\\\\\\\\\\\\n32962\\\\\\\\\\\\\\\\\\\\nDetailed Authentication Information:\\\\\\\\\\\\\\\\\\\\nLogon Process:\\\\\\\\\\\\\\\\\\\\nNtLmSsp\\\\\\\\\\\\\\\\\\\\nAuthentication Package:\\\\\\\\\\\\\\\\\\\\nNTLM\\\\\\\\\\\\\\\\\\\\nTransited Services:\\\\\\\\\\\\\\\\\\\\n\\\\\\\\\\\\\\\\\\\\nPackage Name (NTLM only):\\\\\\\\\\\\\\\\\\\\nNTLM V2\\\\\\\\\\\\\\\\\\\\nKey Length:\\\\\\\\\\\\\\\\\\\\n128\\\\\\\\\\\\\\\\\\\\n\\\\\\\\\\\\\\\\\\\\nThis event is generated when a logon session is created. It is generated on the computer that was accessed.\\\\\\\\\\\\\\\\\\\\n\\\\\\\\\\\\\\\\\\\\nThe subject fields indicate the account on the local system which requested the logon. This is most commonly a service such as the Server service, or a local process such as Winlogon.exe or Services.exe.\\\\\\\\\\\\\\\\\\\\n\\\\\\\\\\\\\\\\\\\\nThe logon type field indicates the kind of logon that occurred. The most common types are 2 (interactive) and 3 (network).\\\\\\\\\\\\\\\\\\\\n\\\\\\\\\\\\\\\\\\\\nThe New Logon fields indicate the account for whom the new logon was created, i.e. the account that was logged on.\\\\\\\\\\\\\\\\\\\\n\\\\\\\\\\\\\\\\\\\\nThe network fields indicate where a remote logon request originated. Workstation name is not always available and may be left blank in some

```

Figure 7 — Wazuh detection for suspicious remote NTLM network logon activity

```
{
  "_index": "wazuh-alerts-4.x-2026.05.25",
  "_id": "xzYfXp4Br749Nx0oTdbO",
  "_version": 1,
  "_score": null,
  "_source": {
    "input": {
      "type": "log"
    },
    "agent": {
      "ip": "10.0.2.15",
      "name": "DESKTOP-HDF09SQ",
      "id": "001"
    },
    "manager": {
      "name": "wazuh-server"
    },
    "data": {
      "win": {
        "eventdata": {
          "subjectLogonId": "0xc8e2cf",
          "subjectUserSid": "S-1-5-21-221611468-1794481200-2977160085-1001",
          "subjectDomainName": "DESKTOP-HDF09SQ",
          "privilegeList": "SeSecurityPrivilege SeBackupPrivilege SeRestorePrivilege SeTakeOwnershipPrivilege"
        }
      }
    }
  }
}
```

```

SeDebugPrivilege  SeSystemEnvironmentPrivilege  SeLoadDriverPrivilege  SeImpersonatePrivilege
SeDelegateSessionUserImpersonatePrivilege",
"subjectUserName": "socadmin"
},
"system": {
"eventID": "4672",
"keywords": "0x8020000000000000",
"providerGuid": "{54849625-5478-4994-a5ba-3e3b0328c30d}",
"level": "0",
"channel": "Security",
"opcode": "0",
"message": "\"Special privileges assigned to new logon.\r\n\r\nSubject:\r\n\tSecurity ID:\t\tS-1-5-21-221611468-1794481200-2977160085-1001\r\n\tAccount Name:\tsocadmin\r\n\tAccount Domain:\t\tDESKTOP-HDF09SQ\r\n\tLogon ID:\t\t0xC8E2CF\r\n\r\nPrivileges:\t\tSeSecurityPrivilege\r\n\t\tSeBackupPrivilege\r\n\t\t\tSeRestorePrivilege\r\n\t\t\t\tSeT
"version": "0",
"systemTime": "2026-05-25T07:52:34.1662186Z",
"eventRecordID": "10390",
"threadID": "3048",
"computer": "DESKTOP-HDF09SQ",
"task": "12548",
"processID": "868",
"severityValue": "AUDIT_SUCCESS",
"providerName": "Microsoft-Windows-Security-Auditing"
}
}
},
"rule": {
"firedtimes": 1,
"mail": false,
"level": 3,
"description": "Special privileges assigned to new logon.",
"groups": [
"windows",
" WEF"
],
"mitre": {
"technique": [
"Domain Policy Modification"
],
"id": [
"T1484"
],
"tactic": [
"Defense Evasion",
"Privilege Escalation"
]
},
"id": "67028"
},
"location": "EventChannel",
"decoder": {
"name": "windows_eventchannel"
},
"id": "1779695564.1907263",
"timestamp": "2026-05-25T07:52:44.903+0000"
},
"fields": {

```

```

"timestamp": [
  "2026-05-25T07:52:44.903Z"
],
},
"highlight": {
  "rule.id": [
    "@opensearch-dashboards-highlighted-field@67028@opensearch-dashboards-highlighted-field@"
  ]
},
"sort": [
  1779695564903
]
}

```

Figure 8 — Wazuh detection for privileged administrative logon activity

TIMELINE + FINDINGS

Attack Timeline

Time	Event
1:22:15 PM	Failed SMB authentication attempt generated
1:22:34 PM	Successful NTLMv2 SMB logon established
1:22:34 PM	Administrative privileges assigned
1:22:34 PM	SMB network session terminated
1:22:44 PM	Wazuh generated failed authentication alert
1:22:44 PM	Wazuh generated remote NTLM authentication alerts

Key Detection Findings

Finding	Impact
Failed SMB authentication detected	Credential guessing indicator
Successful NTLMv2 authentication observed	Remote authenticated access visibility
Administrative privileges assigned	Elevated account usage detected
External source IP identified	Remote source attribution
Logon Type 3 observed	SMB/lateral movement activity
Authentication correlation successful	Improved detection confidence

MOST IMPORTANT IDS TO MENTION

Windows Security Event IDs

Event ID	Meaning
4625	Failed logon
4624	Successful network logon
4672	Special privileges assigned
4634	Logoff

Wazuh Rule IDs

Rule ID	Purpose
60122	Failed authentication detection
92657	Suspicious NTLM remote logon
67028	Privileged logon detection
60137	Windows user logoff

Sysmon Event IDs

Important Note

No major Sysmon process creation telemetry was observed during this simulation because SMB authentication activity primarily occurred through Windows authentication mechanisms and Security Event logging rather than local process execution activity.

The investigation relied mainly on:

- Windows Security Events
- NTLM authentication telemetry
- network logon activity
- Wazuh authentication correlation rules

This highlights the importance of correlating authentication events, source network attribution, and SIEM telemetry during SMB authentication and network logon investigations.

Section — SOC Analyst Assessment

Assessment Category	Value
Severity	Medium
Confidence Level	High
Attack Classification	SMB Authentication Reconnaissance
Source Host	Kali Linux (192.168.56.107)
Target Host	Windows Endpoint (192.168.56.106)
Authentication Method	NTLM Network Logon
Logon Type Observed	Type 3 — Network Logon
Primary Windows Event IDs	4625, 4624, 4634
Primary Wazuh Rule IDs	60122, 92652, 60137
Observed Behavior	Failed SMB authentication followed by successful anonymous NTLM network logon activity
Suspicious Indicators	Remote authentication attempts, NTLM authentication usage, external source IP, SMB enumeration behavior
False Positive Risk	Moderate
Investigation Outcome	Activity confirmed as controlled SMB enumeration performed from a Kali Linux system during SOC detection engineering testing
Analyst Recommendation	Continue monitoring for repeated SMB authentication failures, NTLM-based network logons, anonymous authentication activity, and abnormal SMB enumeration patterns originating from external hosts

Although the activity was generated within a controlled lab environment, the behavioral sequence closely resembled reconnaissance and authentication patterns commonly observed during early-stage lateral movement and SMB enumeration activity in enterprise environments.

Detection confidence increased significantly through correlation between failed authentication attempts, NTLM network logons, source IP attribution, and Wazuh behavioral alerting.

CONCLUSION

Correlated Windows Security Events and Wazuh telemetry provided strong visibility into SMB authentication failures, NTLMv2 network logons, privileged account activity, and remote authentication behavior commonly associated with lateral movement investigations.

Detection confidence increased significantly through behavioral correlation between failed authentication attempts, successful NTLM network logons, administrative privilege assignment, and external source attribution.

The investigation also reinforced the importance of contextual analysis when distinguishing legitimate remote SMB authentication activity from suspicious NTLM-based reconnaissance and authentication behavior during SOC investigations.